

Groups: definition and examples

Applied Algebra
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Big picture. Group theory studies common structure across many mathematical objects.

Key idea.

- Work at an abstract level to reveal essential structure;
- Apply the same results in many different settings;
- Ignore irrelevant details to see the “big picture”.

We begin with **permutations** as concrete examples, and develop intuition before introducing groups formally.

Permutations

A **permutation** of $\{1, \dots, n\}$ is a bijection

$$\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}.$$

It rearranges the elements of the set.

Example. A permutation of $\{1, 2, 3\}$ is given by

$$\sigma(1) = 2, \quad \sigma(2) = 3, \quad \sigma(3) = 1.$$

This sends

$$1, 2, 3 \mapsto 2, 3, 1.$$

Notation

The same permutation can be written in different ways.

Two-line notation:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}.$$

One-line notation:

$$\sigma = (2, 3, 1),$$

meaning $\sigma(i)$ is the i -th entry.

Example

Let

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 1 & 4 & 2 \end{pmatrix}.$$

Then

$$\sigma(1) = 3, \quad \sigma(2) = 5, \quad \sigma(3) = 1, \quad \sigma(4) = 4, \quad \sigma(5) = 2.$$

One-line notation:

$$\sigma = (3, 5, 1, 4, 2).$$

Each number appears exactly once.

The symmetric group

The set of all permutations of $\{1, \dots, n\}$ is denoted by S_n .

Examples.

$$S_1 = \{\text{id}\}, \quad |S_2| = 2.$$

All permutations of $\{1, 2, 3\}$:

$$(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1).$$

In general,

$$|S_n| = n!.$$

Composition of permutations

Let $\sigma, \tau \in S_n$. Their composition is

$$(\sigma \circ \tau)(i) = \sigma(\tau(i)).$$

Apply τ first, then σ .

Example.

$$\sigma = (2, 3, 1), \quad \tau = (3, 1, 2).$$

$$1 \rightarrow 3 \rightarrow 1, \quad 2 \rightarrow 1 \rightarrow 2, \quad 3 \rightarrow 2 \rightarrow 3.$$

$$\sigma \circ \tau = (1, 2, 3).$$

Composition is not commutative

In general,

$$\sigma \circ \tau \neq \tau \circ \sigma.$$

Example.

$$\sigma = (2, 1, 3), \quad \tau = (1, 3, 2).$$

$$\sigma \circ \tau = (2, 3, 1), \quad \tau \circ \sigma = (3, 1, 2).$$

Order matters.

Disjoint permutations

Definition. Let $\pi, \sigma \in S_n$. We say that π and σ are *disjoint* if no element of $\{1, \dots, n\}$ is moved by both permutations; that is,

$$\pi(i) \neq i \Rightarrow \sigma(i) = i, \quad \sigma(i) \neq i \Rightarrow \pi(i) = i.$$

Theorem. If π and σ are disjoint, then they commute:

$$\pi\sigma = \sigma\pi.$$

Idea.

- π and σ act on disjoint sets of elements.
- Each permutation does not affect the action of the other.
- Hence applying them in either order gives the same result.

Identity and inverse

The identity permutation is

$$\text{id} = (1, 2, 3, \dots n).$$

Every $\sigma \in S_n$ has an inverse σ^{-1} such that

$$\sigma \circ \sigma^{-1} = \sigma^{-1} \circ \sigma = \text{id}.$$

Example.

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \Rightarrow \sigma^{-1} = (3, 1, 2).$$

How to find the inverse

Let

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 1 & 4 & 2 \end{pmatrix}.$$

Reverse the arrows:

$$\sigma^{-1}(3) = 1, \quad \sigma^{-1}(5) = 2, \quad \sigma^{-1}(1) = 3, \quad \sigma^{-1}(4) = 4, \quad \sigma^{-1}(2) = 5.$$

Thus

$$\sigma^{-1} = \sigma = (3, 5, 1, 4, 2).$$

Quick check

Let

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 5 & 1 & 3 \end{pmatrix}.$$

- 1 Write σ in one-line notation.
- 2 Compute σ^{-1} .
- 3 Compute σ^2 .

Why new notation?

One-line notation becomes hard to read for large n .

Example:

$$\sigma = (3\ 5\ 1\ 4\ 2).$$

We want a notation that shows the **structure** of a permutation.

Cycle notation

A **cycle** describes how elements move.

Definition. A permutation $\pi \in S_n$ is called a *cycle* if there exist distinct elements $k_1, \dots, k_r \in \{1, \dots, n\}$ such that

$$\pi(k_1) = k_2, \pi(k_2) = k_3, \dots, \pi(k_{r-1}) = k_r, \pi(k_r) = k_1,$$

and $\pi(i) = i$ for all $i \notin \{k_1, \dots, k_r\}$.

The integer r is called the *length* of the cycle. A cycle of length r is called an *r -cycle*. In particular, a 2-cycle is called a *transposition*.

Example:

$$1 \rightarrow 2, \quad 2 \rightarrow 3, \quad 3 \rightarrow 1 \quad \Rightarrow \quad (1 \ 2 \ 3).$$

Example

Let

$$\sigma = (3, 5, 1, 4, 2).$$

Track 1:

$$1 \rightarrow 3 \rightarrow 1 \quad \Rightarrow \quad (1\ 3).$$

Track 2:

$$2 \rightarrow 5 \rightarrow 2 \quad \Rightarrow \quad (2\ 5).$$

Thus

$$\sigma = (1\ 3) \circ (2\ 5) = (1\ 3)(2\ 5).$$

Disjoint cycles

Cycles are **disjoint** if they share no elements.

Example:

$$(1\ 3)(2\ 5)$$

Each element appears in exactly one cycle.

Fixed points are usually omitted.

Another example

Let

$$\sigma = (2, 3, 4, 1, 5).$$

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1 \quad \Rightarrow \quad (1 \ 2 \ 3 \ 4).$$

5 is fixed.

Thus

$$\sigma = (1 \ 2 \ 3 \ 4).$$

Key fact

Every permutation can be written as a product of disjoint cycles.

This decomposition is **unique up to order**.

Example:

$$(1\ 3)(2\ 5) = (2\ 5)(1\ 3).$$

Powers of a cycle

Let

$$\sigma = (1\ 2\ 3).$$

$$\sigma^2 = (1\ 3\ 2), \quad \sigma^3 = \text{id}.$$

A cycle of length k returns to identity after k steps.

Order of a permutation

The **order** of σ is the smallest k such that

$$\sigma^k = \text{id}.$$

Example:

$(1\ 2\ 3)$ has order 3.

$(1\ 2)$ has order 2.

Order of disjoint cycles

Theorem. If σ is a product of disjoint cycles, then

$$\text{order of } \sigma = \text{lcm}(\text{cycle lengths}).$$

Example:

$$\sigma = (1\ 2)(3\ 4\ 5).$$

$$\text{order} = \text{lcm}(2, 3) = 6.$$

Examples

Let

$$\sigma = (1\ 3)(2\ 5).$$

Each cycle has length 2.

$$\text{order} = \text{lcm}(2, 2) = 2.$$

Let

$$\sigma = (1\ 2\ 3\ 4)(5\ 6).$$

Cycle lengths: 4 and 2.

$$\text{order} = \text{lcm}(4, 2) = 4.$$

How to compute order

- 1 Write σ as disjoint cycles.
- 2 Find the length of each cycle.
- 3 Take the least common multiple.

This is much faster than computing powers.

Quick checks

Let

$$\sigma = (4, 1, 3, 2, 5).$$

- 1 Write σ in cycle notation.
- 2 Compute its order.

Let

$$\sigma = (1\ 2\ 3)(4\ 5).$$

- 1 Compute σ^2 .
- 2 Compute σ^3 .
- 3 What is the order?

Transpositions

A **transposition** swaps two elements.

Example:

$$(1\ 2), \quad (3\ 5).$$

All other elements remain fixed.

A transposition has order 2.

Decomposing cycles

Every cycle can be written as a product of transpositions.

Example:

$$(1\ 2\ 3) = (1\ 3)(1\ 2).$$

$$(1\ 2\ 3\ 4) = (1\ 4)(1\ 3)(1\ 2).$$

In general,

$$(a_1\ a_2\ \cdots\ a_k) = (a_1\ a_k) \cdots (a_1\ a_2).$$

Key fact

Every permutation can be written as a product of transpositions.

This decomposition is **not unique**.

Example:

$$(1\ 2\ 3) = (1\ 3)(1\ 2) = (2\ 3)(1\ 3).$$

Even and odd permutations

A permutation is

- **even** if it is a product of an even number of transpositions,
- **odd** if it is a product of an odd number of transpositions.

Key fact. This does not depend on the decomposition.

Define

$$\operatorname{sgn}(\sigma) = \begin{cases} +1 & \text{if } \sigma \text{ is even,} \\ -1 & \text{if } \sigma \text{ is odd.} \end{cases}$$

Example.

$(1\ 2)$ is odd, $(1\ 2\ 3)$ is even.

Sign of a cycle

A k -cycle has sign

$$(-1)^{k-1}.$$

Examples.

$$(1\ 2) \Rightarrow -1, \quad (1\ 2\ 3) \Rightarrow +1,$$

$$(1\ 2\ 3\ 4) \Rightarrow -1.$$

So transpositions are odd, 3-cycles are even, and 4-cycles are odd.

Multiplicativity

For all $\sigma, \tau \in S_n$,

$$\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma) \operatorname{sgn}(\tau).$$

This makes sgn a homomorphism

$$S_n \rightarrow \{\pm 1\}.$$

Alternating group

The set of even permutations in S_n is denoted by

$$A_n = \{\sigma \in S_n : \sigma \text{ is even}\} = \{\sigma \in S_n : \text{sgn}(\sigma) = +1\}.$$

A_n is a “subgroup” of S_n :

Identity: $e \in A_n$.

Closure: $\sigma, \tau \in A_n \implies \sigma\tau \in A_n$.

Inverses: $\sigma \in A_n \implies \sigma^{-1} \in A_n$.

For $n \geq 2$,

$$|A_n| = \frac{1}{2}n!.$$

Reason. Pair each permutation $\sigma \in S_n$ with

$$(12)\sigma.$$

Multiplying by the transposition (12) changes even permutations to odd ones, and odd permutations to even ones. This pairing is reversible because

$$(12)((12)\sigma) = \sigma.$$

Thus the permutations in S_n split into even–odd pairs, so exactly half of the $n!$ permutations are even.

How to compute sign

If σ is a product of disjoint cycles, then

$$\operatorname{sgn}(\sigma) = \prod (\text{signs of cycles}).$$

Method.

- 1 Write σ as disjoint cycles.
- 2 Use $(-1)^{k-1}$ for each cycle.
- 3 Multiply.

This is much faster than counting transpositions.

Examples: computing signs

Example 1. Let

$$\sigma = (1\ 2)(3\ 4\ 5).$$

Cycle lengths are 2 and 3, so

$$\operatorname{sgn}(\sigma) = (-1)^{2-1}(-1)^{3-1} = (-1)(+1) = -1.$$

Thus σ is odd.

Example 2. If

$$\sigma = (3,\ 5,\ 1,\ 4,\ 2),$$

then

$$\sigma = (1\ 3)(2\ 5),$$

so σ is even and

$$\operatorname{sgn}(\sigma) = +1.$$

Quick checks

Check 1. Let

$$\sigma = (1\ 2\ 3)(4\ 5\ 6).$$

- 1 Compute $\text{sgn}(\sigma)$.
- 2 Is σ in A_6 ?

Check 2. Let

$$\sigma = (2, 4, 1, 5, 3).$$

- 1 Write σ in cycle form.
- 2 Compute its sign.

Definition of a group

We have already seen several algebraic structures:

$$(\mathbb{Z}_n, +), \quad G_n, \quad S_n, \quad A_n.$$

They all have:

- a set of objects,
- an operation for combining them,
- an identity element,
- inverses.

A **group** abstracts this common structure.

A **group** is a set G with an operation $*$ satisfying:

① **Closure:** if $g, h \in G$, then $g * h \in G$.

② **Associativity:**

$$(g * h) * k = g * (h * k).$$

③ **Identity:** there is an element $e \in G$ such that

$$e * g = g * e = g.$$

④ **Inverses:** for each $g \in G$, there is $g^{-1} \in G$ such that

$$g * g^{-1} = g^{-1} * g = e.$$

Notation and Abelian groups

We usually write

$$gh$$

instead of

$$g * h.$$

A group is **Abelian** if

$$gh = hg$$

for all $g, h \in G$.

	multiplicative notation	additive notation
operation	gh	$g + h$
identity	e	0
inverse	g^{-1}	$-g$

Identity and inverses are unique

Theorem. In a group G , the identity element is unique, and each element has a unique inverse.

Proof: identity.

Suppose e and f are both identities. Then

$$f = ef = e.$$

Thus $e = f$.

Proof: inverse.

Suppose h and k are both inverses of g . Then

$$h = he = h(gk) = (hg)k = ek = k.$$

Thus $h = k$.

Examples and non-examples

Groups

- $(\mathbb{Z}, +)$
- $(\mathbb{Z}_n, +)$
- $G_n = \mathbb{Z}_n^\times$
- $(\mathbb{R}^\times, \cdot)$
- $(\mathbb{Q}^\times, \cdot)$
- $(\mathbb{C}^\times, \cdot)$

$$\mathbb{R}^\times = \mathbb{R} \setminus \{0\}.$$

Not groups

- $(\mathbb{N}, +)$

No additive inverses.

- $(\mathbb{Z}, \cdot)$

For example, 2 has no inverse in $\mathbb{Z}$.

- $(\mathbb{Z}_n, \cdot)$

The class $[0]_n$ has no inverse.

Example: the group table for $\mathbb{Z}_6$

Consider the group

$$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$$

under addition modulo 6.

+	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

Each entry is computed modulo 6. For example,

$$4 + 5 = 9 \equiv 3 \pmod{6}.$$

Using a group table

For a finite group, the operation can be recorded in a table.

From the table, we can read off important information.

- The identity row and identity column copy the headings.
- The inverse of g is found by locating e in the row or column for g .
- Each group element appears exactly once in every row and column.

For example, in $\mathbb{Z}_6$, the identity is 0, and

$$2 + 4 \equiv 0 \pmod{6},$$

so the inverse of 2 is 4.

Why are there no repeats?

In a group table, each row and column contains every group element exactly once.

Why can there be no repeats in a row?

Suppose two entries in the row for g are equal: $ga = gb$.

Multiplying both sides on the left by g^{-1} , we get

$$g^{-1}(ga) = g^{-1}(gb).$$

Using associativity,

$$(g^{-1}g)a = (g^{-1}g)b,$$

so

$$ea = eb.$$

Therefore,

$$a = b.$$

So two different columns cannot give the same entry in a row.

Examples: permutation groups

The symmetric group

$$S_n = \{\text{all permutations of } \{1, \dots, n\}\}$$

is a group under composition.

The identity is id , and the inverse of a permutation is its inverse function.

Also,

$$A_n = \{\sigma \in S_n : \text{sgn}(\sigma) = +1\}$$

is a group under composition.

Another example inside S_4 is

$$V = \{\text{id}, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}.$$

Examples: matrix groups

Let

$$GL(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) : A \text{ is invertible}\}.$$

Under matrix multiplication, $GL(n, \mathbb{R})$ is a group.

- identity: I_n ,
- inverse of A : A^{-1} ,
- associativity comes from matrix multiplication.

For $n \geq 2$, this group is non-Abelian.

Examples: symmetry groups

A **symmetry** of a figure is a rigid motion that leaves it looking the same.

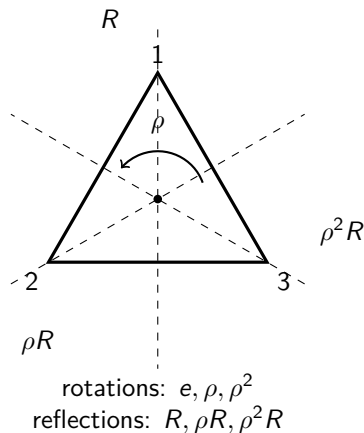
For an equilateral triangle,

$$D_3 = \{e, \rho, \rho^2, R, \rho R, \rho^2 R\}.$$

Here

$$\rho^3 = e, \quad R^2 = e, \quad R\rho = \rho^{-1}R.$$

More generally, D_n is the symmetry group of a regular n -gon with $|D_n| = 2n$.



Quick check

Decide whether each is a group under the stated operation.

- ① $M_2(\mathbb{R})$ under matrix addition
- ② $GL(2, \mathbb{R})$ under matrix addition
- ③ $GL(2, \mathbb{R})$ under matrix multiplication
- ④ The symmetries of a square under composition

For each group, identify the identity element.